

# Static Analysis Report

## Revision

Version 6  
2/13/26 8:16 PM

## SME

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Report: Marwan Abi-Antoun

## Abstract

This document describes the process used to create a static analysis report.

## Group / Owner

DevOps / Information Systems Security Developer

## Motivation

This document is motivated by the need to have early security-related implementation feedback in the development of software for use within safety-critical, cyber-physical systems.

## License

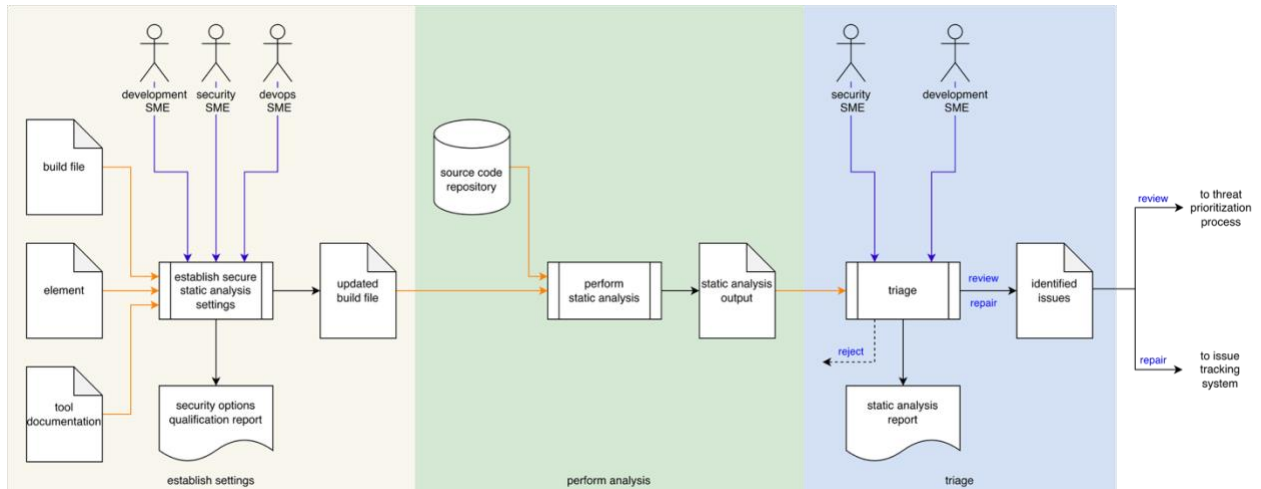
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# Overview

After compiler feedback, static analysis provides the fastest feedback available to developers as to possible security issues within their code.

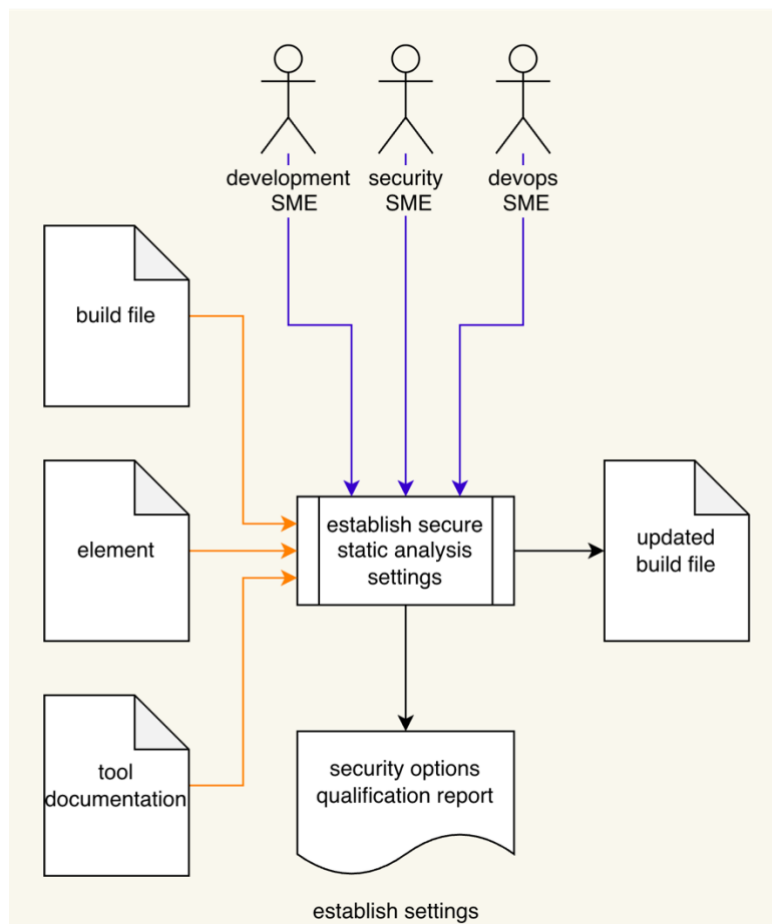
The following diagram illustrates the process to be used:



# Process

## Establish Settings

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| <b>Inputs</b>       | Build file<br>Element<br>Tool documentation                 |
| <b>Outputs</b>      | Updated build file<br>Security options qualification report |
| <b>Participants</b> | Development SME<br>Security SME<br>DevOps SME               |



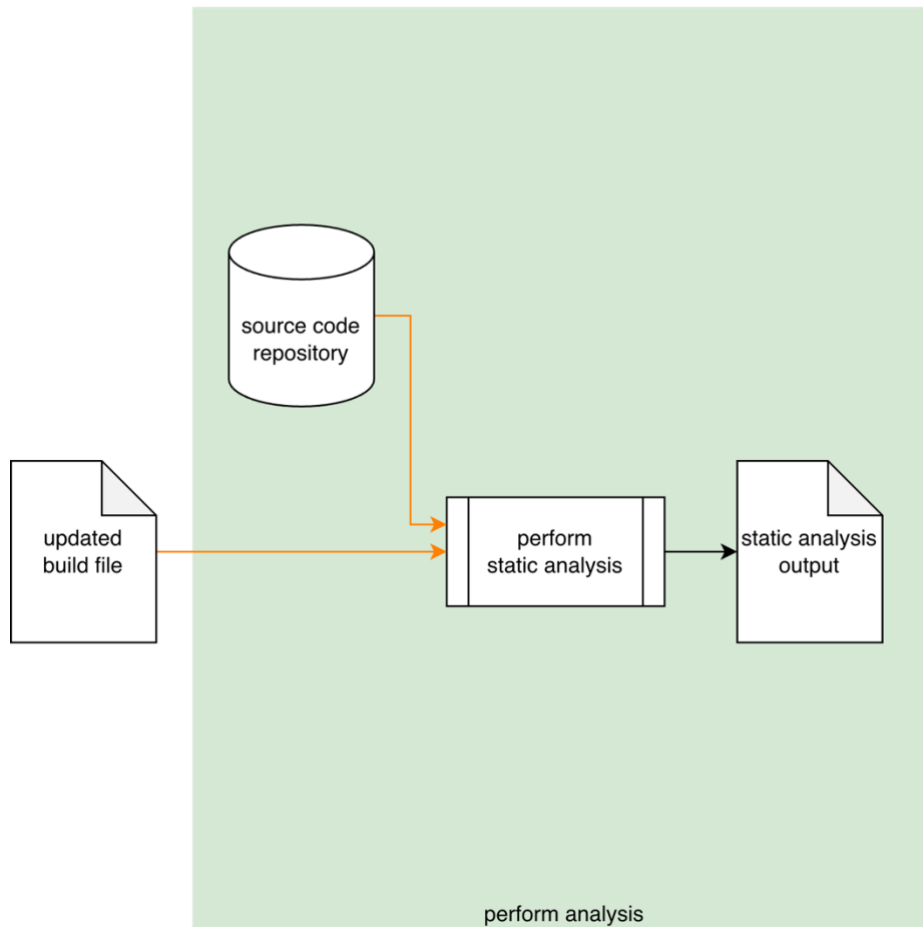
Using the **Element** under consideration, static analysis **Tool Documentation**, and **Build File**; the Development SME, Security SME, and DevOps SME will work together following the **Secure Settings Document** [\[2\]](#) process to determine what element aspect needs to be tested and what

static analysis tool settings are needed to enable that testing. An **Updated Build File** will be produced. A **Security Options Qualifications Report** is generated.

**Note:** The scope of the element may be as small as a single file or as large as the entire project.

## Perform Analysis

|                     |                                              |
|---------------------|----------------------------------------------|
| <b>Inputs</b>       | Updated build file<br>Source code repository |
| <b>Outputs</b>      | Static analysis output                       |
| <b>Participants</b> | none                                         |

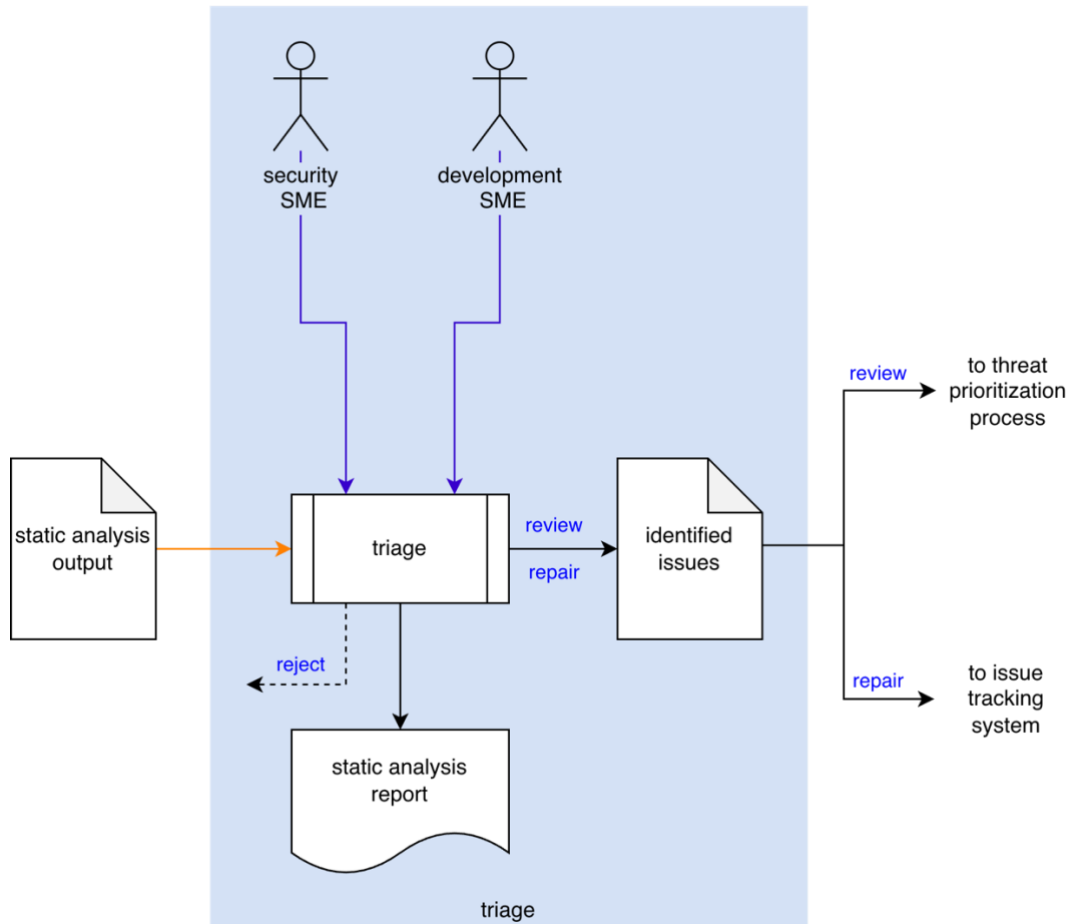


Using the settings from the **Updated Build File**, the **Static Analysis Tool** takes the **Source Code Repository** and performs an analysis on the element. The generated output is captured into **Static Analysis Output**.

**Note:** The scope of the analysis may be as small as a single file or as large as the entire repository.

## Triage

|                     |                                             |
|---------------------|---------------------------------------------|
| <b>Inputs</b>       | Static analysis output                      |
| <b>Outputs</b>      | Identified issues<br>Static analysis report |
| <b>Participants</b> | Security SME<br>Development SME             |



The **Security SME** and **Development SME** review the **Static Analysis Output** to identify any issues needing further investigation. Identified issues are sent to the threat prioritization process [\[6\]](#). A **Static Analysis Report** will be generated.

**Note:** Depending on the nature of an identified issue, it may be entered directly into the issue tracking system.

## Identified Issues

The recommended form of the **Identified Issues** artifact is a Static Analysis Results Interchange Format (**SARIF**) encoded JSON. This document assumes SARIF version 2.1.0 [\[1\]](#) or later.

## Static Analysis Report

The **Static Analysis Report** is recommended to be produced from the **Identified Issues** artifact and should detail the issues exposed by the static analysis.

The report contains one or more analysis runs. Each run includes:

- Description of the tool used
- Description of the analyzed element
- Tool invocation settings
- Results of the analysis

The tool description includes:

- Name
- Version
- URI to tool documentation
- Tool rules

The tool rules (one or more) convey the classes of analysis performed. Each includes:

- ID (unique)
- Name
- Short description of the rule
- Full description of the rule
- URI to rule documentation
- Reference (optional) to associated taxonomy entry (Common Weakness Enumeration, ...)

The analyzed element description (one or more) provides information related to the element under consideration. Each includes:

- URI to analyzed element
- URI to repository the element came from

The analysis results (one or more) describe the issues exposed by the analysis. Each includes:

- Human-readable description
- Location within the element of the issue
- Severity of the issue
- Reference (optional) to the associated taxonomy entry

# References

1. **Static Analysis Results Interchange Format (SARIF) Version 2.1.0**  
<https://docs.oasis-open.org/sarif/sarif/v2.1.0/os/sarif-v2.1.0-os.pdf>
2. **Secure Settings Document** (AVCDL secondary document)
3. **Fuzz Testing Report** (AVCDL secondary document)
4. **Dynamic Analysis Report** (AVCDL secondary document)
5. **Security Options Qualification Report** (AVCDL tertiary document)
6. **Threat Prioritization Plan** (AVCDL secondary document)